Guided Tutorial: Introduction to Machine Learning

Supervised & Unsupervised Learning using Python & Scikit-Learn

Learning Objectives

By the end of this tutorial, you should be able to:

- Understand key ML concepts (**supervised, unsupervised, regression, classification, clustering**)
- Train ML models using sample datasets (Iris, house prices)
- Evaluate model accuracy using metrics like **MSE**, **accuracy**, **confusion matrix**
- Visualize and interpret model outputs

Key Ideas to Recall

Before we start, let's review some essential terms used in Machine Learning.

Term	Meaning (in simple words)	Example
Data (Features)	The input variables we use to make predictions	House size, number of rooms
Label (Target)	The output we want to predict	House price
Model	A rule or pattern learned from data	$y = m \times x + b$
Training	Teaching the model using known examples	Past house prices
Testing	Checking how well the model works on unseen data	New house prices
Prediction	The result the model produces	Estimated price
Error / Loss	How far off the prediction is from the real value	Actual - Predicted

Supervised vs. Unsupervised Learning

Machine Learning algorithms can be divided into two main types based on the availability of labels

Type	Description	Example
Supervised Learning	Model learns from labeled data (answers are known).	Predicting house prices
Unsupervised Learning	Model finds patterns in unlabeled data.	Grouping customers by behavior

Linear Regression (Supervised Learning)

Linear Regression finds a straight-line relationship between an input (x) and output (y): y = m*x + b

Example: Predicting House Price from Area (sq ft)

Square Feet	Price ($)
1000	180,000
1500	240,000
2000	300,000

Question 1: Does the price increase or decrease with area?

Answer: _____

Steps to train a Linear Regression model:

1. Start with random values for m and b.
2. Predict prices using y = m*x + b.
3. Compute the error (difference between predicted and actual values).
4. Use Gradient Descent to adjust m and b.
5. Repeat until the error is minimal.

Formula for Mean Squared Error (MSE): MSE = (1/n) Σ (y_true - y_pred)^2

Practical Steps

Step 1: Setup Environment

pip install numpy matplotlib scikit-learn

Step 2: Create a Python Script

```
import numpy as np

from sklearn.linear_model import LinearRegression

import matplotlib.pyplot as plt

square_feet = np.array([1400, 1600, 1700, 1875, 1100, 1550, 2350, 2450, 1425, 1700])

house_prices = np.array([245000, 312000, 279000, 308000, 199000, 219000, 405000, 324000, 319000, 255000])

square_feet = square_feet.reshape(-1, 1)

model = LinearRegression()

model.fit(square_feet, house_prices)

predicted = model.predict(square_feet)

plt.scatter(square_feet, house_prices)

plt.plot(square_feet, predicted)

plt.xlabel("Square Feet")

plt.ylabel("House Price ($)")

plt.show()
```

Step 3: Activity

📌 **Predict price for 2200 sq.ft using:**

```
print(model.predict([[2200]]))
```

Reflection Questions

1. What is the difference between the **independent variable** and the **dependent variable**?

2. Would this model work for predicting *classification* values? Why?

Decision Trees (Supervised Learning)

A Decision Tree makes predictions by asking a sequence of questions based on input features.

Example 1 : Fruit Classification

Diameter (cm)	Color	Fruit
3	Red	Strawberry
6	Orange	Orange
8	Green	Apple

If Diameter < 4 cm → Strawberry; Else if Color = Orange → Orange; Else → Apple

🟩 Question: For diameter = 7 cm and color = orange, predict the fruit:

👉 Answer: _____

Example 2: Predict a label (class) like species using the Iris dataset

Step-by-Step Code

```
from sklearn.datasets import load_iris
from sklearn import tree
import matplotlib.pyplot as plt
iris = load_iris()
X, y = iris.data, iris.target
classifier = tree.DecisionTreeClassifier()
classifier.fit(X, y)
tree.plot_tree(classifier)
plt.show()
```

Activity

Modify the code to predict the class for:

```
Sepal length = 6.3
Sepal width  = 3.2
Petal length = 1.4
Petal width  = 2.0
print(classifier.predict([[6.3, 3.2, 1.4, 2.0]]))
```

Discussion

- What determines the splits in a decision tree?(*Hint: Gini Index, page 35*)
- What is a **leaf node**?

Unsupervised Learning (Clustering)

K-Means groups data points into K clusters based on similarity.

Steps in the K-Means algorithm:

- Choose K (number of clusters).
- Randomly select K centroids.
- Assign each data point to its nearest centroid.
- Recalculate centroids.
- Repeat until centroids stabilise.

Practical Implementation

```
from sklearn.cluster import KMeans

from sklearn.datasets import load_iris

import matplotlib.pyplot as plt

iris = load_iris()

X = iris.data

kmeans = KMeans(n_clusters=3, random_state=0)

labels = kmeans.fit_predict(X)

plt.scatter(X[:, 0], X[:, 1], c=labels)

plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], s=200)

plt.xlabel("Feature 1")

plt.ylabel("Feature 2")

plt.show()
```

Questions

1. What differences do you observe between clustering vs classification?
2. What determines the number of clusters **K**?